

Single Agent 3 MCP tools vs Multiagent multistep interaction

Comparing both approaches using a 1 shot prompt for holiday planning

Here are the prompts

[One agent system](#)

[A2A-multi-agent system](#)

Overall Objective

1. Query Google Cal for available dates based off user's {duration}
2. Query Wikipedia for where to go / things to do in {location}
3. Find airbnb based off user's {budget}
4. Make a plan for the holiday
5. Update Google Cal with the plan
6. Return the plan to the user in md format

Both systems are on an "equivalent" prompt, neither has a harder task.

eg. the multi agent system can likely do a better 6, but was not given a better format for 6, but instead was required to give the same format.

It can also likely do a better research job on wikipedia but also was not asked to do intensive research etc.

- Sub-agents were given simple prompts to basically just "do the task"

Overall Observation with single Agent multiple MCP tools

Even when clearly defined in system prompts it misses out MCP tools eg "sequential thinking" which helps it prepare and go through its thoughts

- Because of inconsistent behaviour Token use is variable
has tendency to make mistakes even on tool calls, this has caused cost to balloon up to ~0.50 USD per request
Tendency to get distracted with multiple tools
- Has recommended AirBnbs not even in the target location
(It has given me Airbnbs in Phuket multiple times)
- Asked for additional information to make reservations? (Hallucination)

Harder to shape consistent tool calls eg. first call tool 1, do this then tool 2 do this
(however this might be a prompt engineering potential fix)

On best case Scenario (rare)

- around 0.13 USD per call, still imperfect information
- About 55 Seconds for a full response
- Quite a well populated calendar output (but still lacks alot of detail)

Overall Observation with Multi Agent one MCP tool each

Development (planning how to prompt each agent, how each agent should interact with each other) was relatively easy, in fact I simply "translated" the one-shot prompt from the single agent into a multi-agent by asking chatgpt and it was a decent starting point but needed extra input from me to fine tune each agent.

Main struggle was the orchestrator agent, Given the multistage interaction with multiple invoke calls, the agent often doesn't know which state it is in.

The agent goes through our own managed context to try to figure out where it is which is definitely suboptimal and has led to repeated tool calls which slow down.

- The current interaction has 5 distinct states, with a total min of 6 invokes(), 4 delegation tasks (one is a parallel delegation)
State information is hard for the model to extract based off purely context, states 1-3 are very well orchestrated but model gets lost after
Currently it forgets to upload to google cal, I just tell it to with an extra message
(State 4) Solution should be something like lang-graph has, a scratch pad so the agent

can write stuff down ie. place to preserve stateful information in multistep interactions so it does not need to infer from context

Tendency for going "off course" or distracted was low, the only issue i had when building and fine tuning the prompt was the states, where the agent will get lost.

- Information I received was mostly reliable, with no strange returns or hallucinations by the agent

There was one instance only where i got an airbnb elsewhere, it was after repeated calls and the orchestrator getting lost, i would guess context clouded out the location requirement of Bangkok

Tool use in general was controlled and quite predictable and didn't cause problems

- Sequential Thinking was interesting in the orchestrator as it allows us to control how long we want the time to be, we can "Turn off thinking" for the orchestrator to be faster

Average Case Scenario

- 90s for complete run
- costs 0.017 USD for full orchestration
- Good markdown return with good airbnb selections
- Missing Google Cal update, but fully populated once I tell orchestrator to add it to my calendar

Overall Thoughts

Initial setup and prompting time for multi agents took quite abit more, but it was also because it was much easier to debug, I can tell how and where the agent was getting lost, which state it was getting lost in etc, hence i could actually fine tune and debug. It was less of a black box and much more predictable than the single agent version.

In comparison when I was trying to debug the single agent system, i really didnt know what to do, the variance in the types of problems was quite high and hard to trace for.

The context management for our A2A system is difficult for stateful interactions, we likely need to provide it a stateful tool to keep track of where it is in complex tasks instead of just inferring from context and previous replies from delegation tasks

Speed wise, in simple requests, I think a simple agent might win out as it doesn't need to delegate, but at the same time there is a high likelihood of it not working as well for harder to use MCPs eg. notion. specialized agents per tool allow us to give tool based prompts to the agent on what to use and when, which might overwhelm a single agent system.

Full Response data

Note that for Single Agent runs, testing was not done excessively because of excessive cost from failed tool use calls etc, it was also very unpredictable in general

Test 1 (Single Agent many MCP Tools)

This one represents a better job being done by the agent

Token Usage:

- Input tokens: 216235
- Output tokens: 2239
- Total tokens: 218474
- Total requests: 9

Bangkok – 7 Days Relaxing Holiday

Dates: 2025-07-17 → 2025-07-24 **Travellers:** 2

Overview

- Explore the majestic Grand Palace and Wat Phra Kaew
- Relax at Lumphini Park and enjoy nature
- Shop and dine at the famous Chatuchak Weekend Market
- Experience the rich culture at Jim Thompson House
- Scenic cruise along the Chao Phraya River

Accommodation: *Prime location in Old Town, Fast WiFi, 50 SQM*, [link](#)

Day 1 – Thursday 17 Jul

Morning

- Arrive in Bangkok. Check-in at your Airbnb.

Afternoon

- Rest and explore local eateries nearby.

Evening

- Visit a local market for dinner.
-

Day 2 – Friday 18 Jul

Morning

- **Visit Grand Palace**
Explore the majestic Grand Palace and Wat Phra Kaew. Entry fee: ~500 THB.
[More info here](#)

Afternoon

- **Relax at Lumpini Park**
Enjoy a relaxing afternoon. Rent a paddle boat or have a picnic.
[More info here](#)

Evening

- Dinner at a local restaurant.

Day 3 – Saturday 19 Jul

Morning

- **Explore Chatuchak Weekend Market**
Shop for souvenirs and savor street food.
[More info here](#)

Afternoon

- Continue exploring the market.

Evening

- **Dinner at a rooftop restaurant**
Enjoy magnificent views of the Bangkok skyline.
-

Day 4 – Sunday 20 Jul

Morning

- **Visit Jim Thompson House**
Explore the traditional Thai house and learn about silk weaving. Entry fee: ~200 THB.
[More info here](#)

Afternoon

- **Cruise on Chao Phraya River**
Enjoy a scenic boat ride. Book in advance.
[More info here](#)

Evening

- Dinner and free time.

Day 5 – Monday 21 Jul

Morning

- **Explore Bangkok Old Town**

Wander through historic neighborhoods, visit local food stalls.

[More info here](#)

Afternoon

- Leisure time in the area.

Evening

- Dinner at a traditional Thai restaurant.

Day 6 – Tuesday 22 Jul

Morning

- Day at leisure. Option to revisit favorite spots or relax at the Airbnb.

Afternoon

- Optional cooking class or spa treatment.

Evening

- Dinner at a restaurant of your choice.

Day 7 – Wednesday 23 Jul

Morning

- Free morning for last-minute shopping or exploring.

Afternoon

- Final lunch in Bangkok.

Evening

- **Check-out Airbnb**
Head to the airport for your departure.

Key Experiences

Experience	When	Link
Visit Grand Palace	Day 2 AM	Grand Palace
Relax at Lumpini Park	Day 2 PM	Lumphini Park
Explore Chatuchak Market	Day 3 AM	Chatuchak Market
Visit Jim Thompson House	Day 4 AM	Jim Thompson House
Cruise on Chao Phraya River	Day 4 PM	Chao Phraya
Explore Bangkok Old Town	Day 5 AM	Bangkok

Practical Notes

- Currency: Thai Baht (THB)
- Plug Type: Type A, B, or C
- Language: Thai is widely spoken; English is commonly understood in tourist areas.
- Travel insurance is recommended.
- Download Google Maps for offline access.

Output to google Cal

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Check-in at Airbnb

Thursday, July 17 · 3:00 – 4:00pm

🔗

 Invite via link

☰

Check-in to the Airbnb property for the relaxing stay:
[Prime location in Old Town, Fast WiFi, 50 SQM]
(<https://www.airbnb.com/rooms/1428007797034678318>)

🔔

 30 minutes before

📅

 HONGYI

Airbnb location was not even in Bangkok

13	14	15	16	17	18	19
				1am Bangkok Trip		
				• 3pm Check-in at Airbnb	• 10am Visit Grand Palace • 3pm Relax at Lumpini P	• 11am Explore Chatuchak • 8pm Dinner at a rooftop
20	21	22	23	24	25	26
Bangkok Trip						
• 11am Visit Jim Thompson • 3pm Cruise on Chao Phr.	• 11am Explore Bangkok O			• 12pm Check-out Airbnb		

✎ Test 2 (1 Agent multiple MCP Tools)

Here token use is much less, the agent also did much less and did a lazy job with scheduling on google cal

📊 Token Usage:

• Input tokens: 137882

- Output tokens: 1409
- Total tokens: 139291
- Total requests: 5

Bangkok – 7 days relaxing trip

Dates: 2025-07-17 → 2025-07-24 **Travellers:** 2

Overview

- Explore iconic landmarks like the Grand Palace and Wat Phra Kaew.
 - Relax in lush green spaces like Lumpini Park.
 - Savor authentic street food in vibrant neighborhoods.
 - Enjoy river cruises to experience the city's waterways.
 - Indulge in a mix of local culture and modern comforts.
 - Accommodation: *Sathorn's Urban Oasis: 1BR with Pool, Gym & Yoga*, [link](#)
-

Day 1 – Thursday 17 July

Morning

- Arrive in Bangkok. Check into your accommodation.
- Refresh and relax after your journey.

Afternoon

- Visit the **Grand Palace**, the former royal residence.
- Explore **Wat Phra Kaew**, home to the Emerald Buddha.

Evening

- Dinner at a local street food stall in the nearby area.

Day 2 – Friday 18 July

Morning

- Stroll through **Lumphini Park**. Enjoy a canoe ride on the lake.

Afternoon

- Lunch at a café nearby, followed by a visit to **Jim Thompson House**.

Evening

- Dinner at a riverside restaurant with views of the river.

Day 3 – Saturday 19 July

Morning

- Head to **Chatuchak Weekend Market** for local shopping and snacks.

Afternoon

- Enjoy lunch at the market and continue browsing its many stalls.

Evening

- Explore the lively atmosphere of **Khao San Road**, dine in local restaurants.

Day 4 – Sunday 20 July

Morning

- Relax with a Thai massage at a reputable spa.

Afternoon

- Lunch, followed by a visit to **Wat Arun**, a stunning riverside temple.

Evening

- Enjoy a leisurely dinner cruise along the Chao Phraya River.

Day 5 – Monday 21 July

Morning

- Visit **Bangkok Art and Culture Centre** for contemporary art exhibits.

Afternoon

- Lunch at a café, then explore the nearby **Siam Square** for shopping.

Evening

- Dinner at a rooftop restaurant with panoramic views of the city.

Day 6 – Tuesday 22 July

Morning

- Take a day trip to **Ayutthaya**, the ancient capital (book a tour).

Afternoon

- Explore the historical temples and ruins.

Evening

- Return to Bangkok for a quiet dinner at a local restaurant.

Day 7 – Wednesday 23 July

Morning

- Explore the **Erawan Shrine** and surrounding shopping areas.

Afternoon

- Last-minute shopping at **Terminal 21** mall.

Evening

- Enjoy a farewell dinner at a renowned Thai restaurant.

Day 8 – Thursday 24 July

Morning

- Check out and prepare for departure.
-

Key Experiences

Experience	When	Link
Grand Palace & Wat Phra Kaew	Day 1 AM	Wikipedia
Lumphini Park	Day 2 AM	Wikipedia
Chatuchak Weekend Market	Day 3 AM	Wikipedia
Wat Arun	Day 4 PM	Wikipedia
Dinner Cruise	Day 4 PM	-

Practical Notes

- **Currency:** Thai Baht (THB)
- **Plug Type:** Type A, B, and C
- **Language:** Thai
- **Travel Insurance Recommendation:** Highly advised to cover health and travel disruptions.

- **Download Offline Maps:** Necessary for navigating the city.

Trial 2 output to google cal



Bangkok Trip

July 17, 2025, 12:00am – July 24, 2025, 11:59pm



Invite via link



Bangkok, Thailand



Trip to Bangkok for relaxation and exploration.

Accommodations:

- [Sathorn's Urban Oasis: 1BR with Pool, Gym & Yoga]
(<https://www.airbnb.com/rooms/1309394937494731137>): 5.0 stars from 7 reviews, Price: 7500฿ for 7 nights.
- [Nomad's Palace]
(<https://www.airbnb.com/rooms/1458585823277268138>): Price after discount: 4474฿ for 7 nights.
- [Studio in Sukhumvit/Free Pool/Professional Service]
(<https://www.airbnb.com/rooms/1335251565832319665>): 4.78 stars from 23 reviews, Price: 5138฿ for 7 nights.

Activities to consider:

- Exploring the Grand Palace and Wat Phra Kaew
- Relaxing in Lumpini Park
- Enjoying local cuisine in Khao San Road



30 minutes before



HONGYI

Instead of multiple events like in trial 1, this only created one event, likely cause of the missing of the Sequential thinking MCP tool and planning steps

I only have the token count for this one, but its token use is significantly less, note that the return format is very consistent, this quality is observed across all trails and the agent has given me a good response 95-99% of the time

Total Token Count:

- **Prompt tokens:** $2403 + 2691 + 3160 + 4196 + 3498 + 2864 + 9816 + 988 = 29,616$
- **Completion tokens:** $109 + 121 + 165 + 357 + 83 + 13 + 509 + 156 = 1,513$

Bangkok – 7-Day Relaxing Holiday

Dates: 2025-07-19 → 2025-07-25

Travellers: 2 (Couple)

Budget: ~3,000 THB per person per night

Accommodation (Top 3 Picks)

1. [Modern 1Bedroom/500 Mbps WiFi MRT Sutthisarn \(201\)](#) – ฿5,598 for 6 nights
2. [Sukhumvit 81 BTS Onnut 2minwalk - GP House Express](#) – ฿3,284 for 6 nights
3. [New! Gateway/ Bangkok University 1BR 2PPL near BTS](#) – ฿9,011 for 6 nights

Itinerary Overview

Day 1 – Saturday 19 Jul

- **Morning:** Visit [Lumphini Park](#) for a peaceful walk or paddleboat ride
- **Lunch:** Thai cuisine at a local restaurant
- **Afternoon:** Rest at accommodation
- **Evening:** Dinner at a rooftop bar with city view

Day 2 – Sunday 20 Jul

- **Morning:** Meditation or yoga class at a local wellness center
 - **Lunch:** Light meal at a cozy café
 - **Afternoon:** [Boat ride along the Chao Phraya River](#)
 - **Evening:** Dinner at a traditional Thai restaurant
-

Day 3 – Monday 21 Jul

- **Morning:** Traditional Thai massage at a reputable spa
 - **Lunch:** Casual street dining
 - **Afternoon:** Explore [Wat Arun \(Temple of Dawn\)](#)
 - **Evening:** Relax and enjoy a light dinner
-

Day 4 – Tuesday 22 Jul

- **Morning:** Leisurely breakfast and rest
 - **Lunch:** Street food from a local market
 - **Afternoon:** Visit a cultural museum or art gallery
 - **Evening:** Dinner with live traditional music
-

Day 5 – Wednesday 23 Jul

- **Morning:** Visit a local temple or cultural site
 - **Lunch:** Meal at a peaceful café
 - **Afternoon:** Relax at your stay or nearby park
 - **Evening:** Cultural show or traditional performance
-

Day 6 – Thursday 24 Jul

- **Morning:** Thai cooking class
 - **Lunch:** Enjoy your own cooked meal
 - **Afternoon:** Free time for shopping or exploring
 - **Evening:** Dinner at a highly rated restaurant
-

Day 7 – Friday 25 Jul

- **Morning:** Final spa treatment or massage
 - **Lunch:** Light meal
 - **Afternoon:** Stroll in Lumpini Park or riverside
 - **Evening:** Farewell dinner at a rooftop bar
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Extra Things to Note

- Public transport like BTS Skytrain or riverboats are recommended for most activities.
 - Book spa appointments, cooking classes, and cultural shows in advance.
 - Always confirm Airbnb listings and cancellation policies before booking.
-

Google cal output

13	14	15	16	17	18	19
						• 4pm Check-in at Airbnb:
20	21	22	23	24	25	26
• 10am Visit Lumpini Park	• 3pm Traditional Thai Mas	• 11am Boat Ride on Chao	• 10am Explore Wat Arun (	• 4pm Tea Ceremony at Tr	• 9am Meditation or Yoga : • 4pm Relax or Revisit Fav	

Sub-Agent consistently added the complete calendar into notion (no shortcuts of simply adding one event etc)

Similar across the board here, everything seems reasonable

Bangkok – 7-Day Relaxing Holiday

 **Dates:** 24 July – 30 July 2025

 **Travellers:** 2 adults

 **Budget:** ≤ ฿3,000 per person per night

Top Airbnb Stays (Entire Place, Within Budget)

1. [Cozy room near BTS – Iconsiam B203](#)
 - Price: ฿4,656 for 6 nights | ★ 5.0 (10 reviews)
2. [At BTS metro room w/ free WiFi – Bangkok 2/1](#)
 - Price: ฿3,732 for 6 nights (after discounts) | ★ 4.9 (10 reviews)
3. [16Rama9 High-End Studio – Infinity Pool/Gym/Sky Garden](#)
 - Price: ฿5,729 for 6 nights | ★ 5.0 (5 reviews)

Daily Itinerary

Day 1 – Thursday, 24 July

 **Morning:** Relaxing walk & paddle boating at [Lumphini Park](#)

 **Afternoon:** Visit cozy cafes in [Ari Neighborhood](#)

 **Evening:** Dinner near your accommodation & unwind

Day 2 – Friday, 25 July

 **Morning:** Cycle through [Bang Krachao](#) – “the green lung of Bangkok”

 **Afternoon:** Chill at [Chatuchak Park](#)

 **Evening:** Dinner & walk in [Thonglor](#)

Day 3 – Saturday, 26 July

 **Morning:** Local markets or downtime at accommodation

 **Afternoon:** [Chao Phraya River Cruise](#) – scenic relaxation

 **Evening:** Dinner by the riverside

Day 4 – Sunday, 27 July

 **Morning:** Lazy breakfast & morning rest

 **Afternoon:** Explore more in Ari – cafes, shops

 **Evening:** Relax at your Airbnb

Day 5 – Monday, 28 July

 **Morning:** Stroll in Lumpini Park or nearby

 **Afternoon:** More time at Chatuchak Park or nearby museum

 **Evening:** Dinner in Thonglor

Day 6 – Tuesday, 29 July

 **Morning:** Revisit Bang Krachao (cycling or walk)

 **Afternoon:** Shopping or enjoy a Thai spa

 **Evening:** Pack up and relax

Day 7 – Wednesday, 30 July



Morning: Pack and check out



Afternoon: Last-minute shopping or walk



Evening: Prepare for departure

✨ Optional Activities

- Visit local art galleries
- Quiet meditation at local temples
- Traditional Thai massage session

Would you like me to **create calendar events** for this itinerary, including activity times and Airbnb links? Just let me know — or if you'd like any edits, I'm happy to adjust!